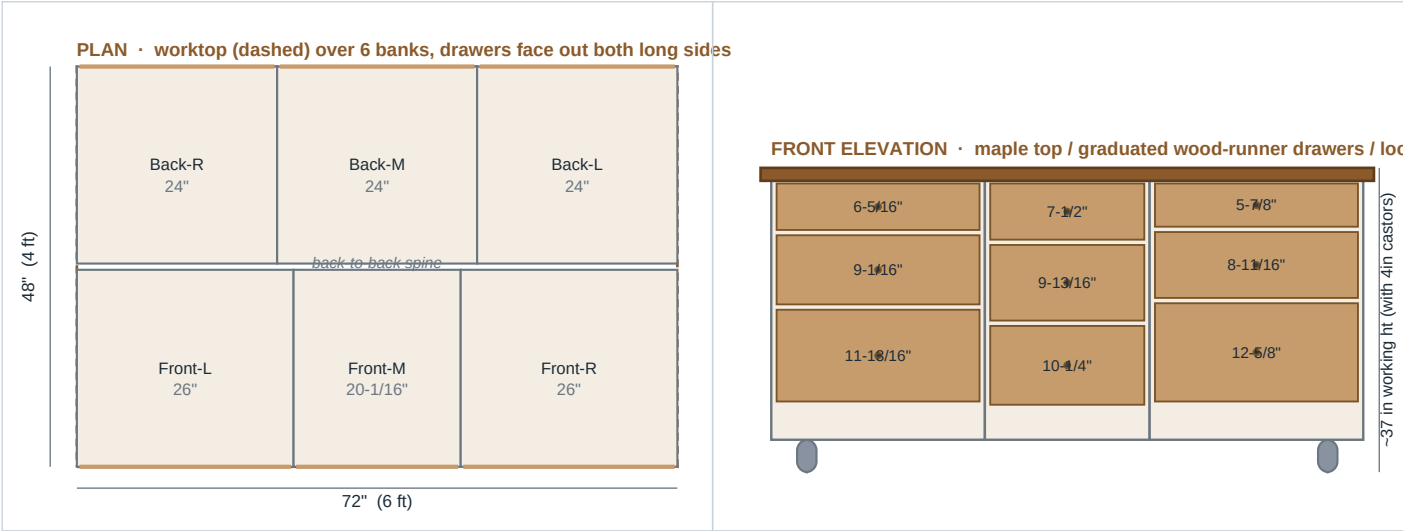


Mobile Storage Workbench

Double-sided mobile island · storage-focused · 4 ft × 6 ft

OVERALL	72 × 48" (6 × 4 ft)	WORKING HT	≈36–37 in (on castors)
DRAWERS	18 · graduated · 3 tiers × 6 banks	RUNNERS	wood-on-wood (no metal slides)
WORKTOP	1-9/16" laminated hard maple	CARCASS	11/16" birch plywood
MOBILITY	4 × heavy-duty locking castors	EST. COST	\$2,564 materials



Construction & specification

Carcass	11/16" birch plywood; sides house the bottom in a 3/8" dado.
Drawer boxes	1/2" ply, dovetailed corners (tails on the sides — can't pull off), 1/4" captured bottom.
Drawer runners	Hardwood side runners screwed to the carcass; drawer sides grooved to ride them. No metal slides, no slide drilling.
Worktop	1-9/16" laminated hard maple, one continuous slab flush to the 4×6 footprint.
Mobility	4 heavy-duty locking swivel castors (rate ≥200 kg each) on corner blocks / base rail.
Stability	Low, wide, heavy double-sided island; wood runners aren't full-extension, so no wall anti-tip (deliberate).

Bank schedule

Bank	Width (in)	Drawer fronts, top → bottom (in)
Front-L	26	6-5/16 · 9-1/16 · 11-13/16 (sum 27-3/16)
Front-M	20-1/16	7-1/2 · 9-13/16 · 10-1/4 (sum 27-9/16)
Front-R	26	5-7/8 · 8-11/16 · 12-5/8 (sum 27-3/16)
Back-R	24	7-7/8 · 9-7/16 · 9-13/16 (sum 27-3/16)
Back-M	24	5-7/8 · 8-1/4 · 13 (sum 27-3/16)
Back-L	24	7-1/16 · 9-7/16 · 10-5/8 (sum 27-3/16)

Cut-list summary (full per-part list in cutlist_imperial.csv)

Part	Qty	Typical size (in)	Material
Drawer box side	36	22-3/8 x 4-5/16 x 1/2 (+14 sizes)	drawer box
Side	12	31-1/2 x 23-5/8 x 11/16	sheet
Bottom	6	18-5/8 x 23-3/8 x 11/16 (+3 sizes)	sheet
Drawer box bottom	18	19-1/16 x 22-3/8 x 1/4 (+3 sizes)	back panel
Top stretcher	12	18-5/8 x 3-1/8 x 11/16 (+3 sizes)	sheet
Back	6	30-13/16 x 18-5/8 x 1/4 (+3 sizes)	back panel
Drawer box front/back	36	18-1/8 x 5-7/8 x 1/2 (+16 sizes)	drawer box
Run countertop	1	72 x 48 x 1-9/16	countertop
Drawer front	18	19-13/16 x 7-1/2 x 11/16 (+16 sizes)	door/front
Drawer wood runner	36	22-3/8 x 1-3/16 x 11/16	solid panel

Parts	181 (115 unique)	Sheet goods	~30.1 m² (324 ft²)
Hardwood runners	36 strips	Edge banding	164 ft
Drilling	108 holes (all wood-runner screws — 0 slide holes)		
Est. build time	~33 h (advanced)	Est. materials	\$2,563.68

Generated by the Woodworking AI pipeline (validator + geometry critic passed, 0 interferences). Full reports alongside: cutlist · drilling · joinery · cutlayout.dxf.
Dimensions overall in in unless noted.

Materials to buy (bill of materials)

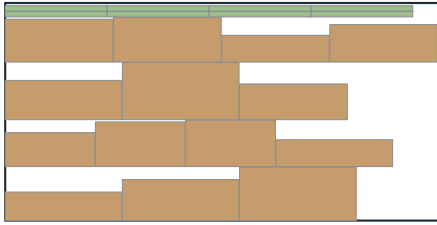
Category	Item	Qty	Unit	Buy cost
Sheet goods	Birch plywood — 1/2", 96-1/16 × 48-1/16" sheet	3	sheet	
Sheet goods	Birch plywood — 1/4", 96-1/16 × 48-1/16" sheet	4	sheet	
Sheet goods	Birch plywood — 11/16", 96-1/16 × 48-1/16" sheet	5	sheet	
Worktop	Hard maple worktop slab — glue-up ≈72-1/16 × 48-1/16 × 1-9/16"	1	slab	
Lumber	Hardwood for drawer runners (Birch sheet)	4.7	bd ft	\$32.34
Hardware	Drawer pull	18	ea	\$63.00
Hardware	Assembly screw 4×35	24	ea	\$2.40
Hardware	Back panel screw 4×16	60	ea	—
Hardware	Edge banding	6	ea	—
Edge banding	Edge banding — sheet	164	ft	\$74.94
Consumable	Wood glue (PVA)			\$96.00
Consumable	Sandpaper (assorted grits)			\$5.40

Sheet count is from the nesting layout (12 plywood sheets + 1 maple slab, 78% yield). Labour (~\$1,131, ~17 h) and consumables are shown for a full quote; materials + hardware to buy total roughly \$1,433.

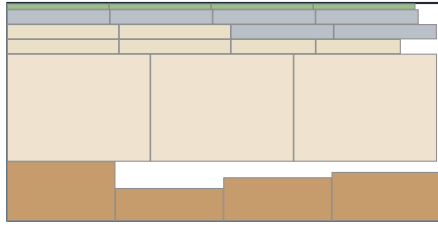
Cut sheets (nesting layout on 96-1/16 × 48-1/16" stock)

■ Carcass side/bottom
 ■ Drawer front
 ■ Drawer box
 ■ Wood runner
 ■ Back panel
 ■ Worktop

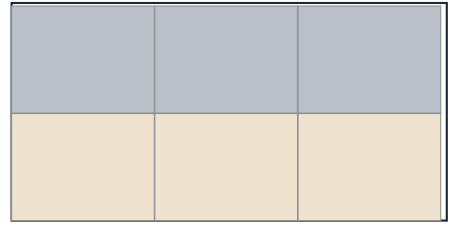
Sheet 1 — 11/16" birch ply 68% used · 22 parts



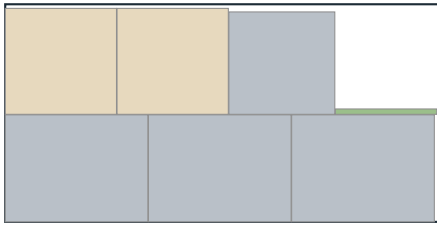
Sheet 2 — 11/16" birch ply 90% used · 23 parts



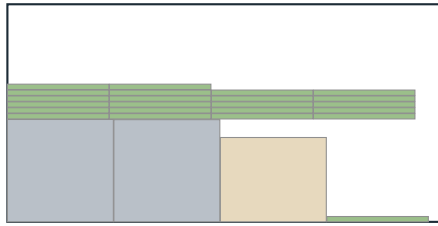
Sheet 3 — 11/16" birch ply 97% used · 6 parts



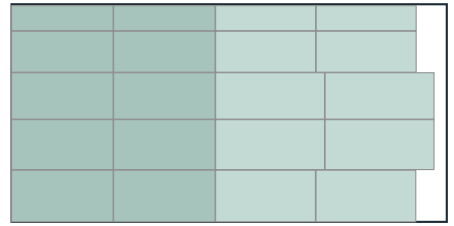
Sheet 4 — 11/16" birch ply 85% used · 7 parts



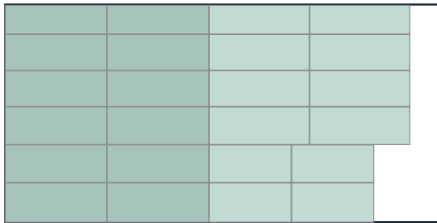
Sheet 5 — 11/16" birch ply 46% used · 26 parts



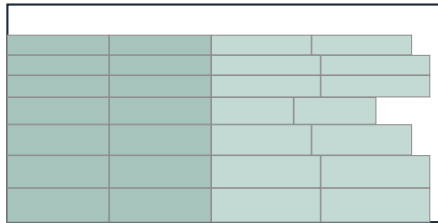
Sheet 6 — 1/2" birch ply 93% used · 20 parts



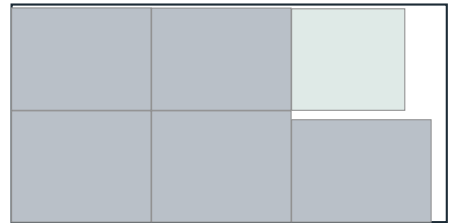
Sheet 7 — 1/2" birch ply 88% used · 24 parts



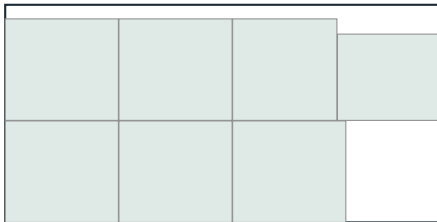
Sheet 8 — 1/2" birch ply 79% used · 28 parts



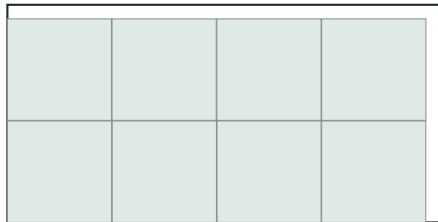
Sheet 9 — 1/4" birch ply 90% used · 6 parts



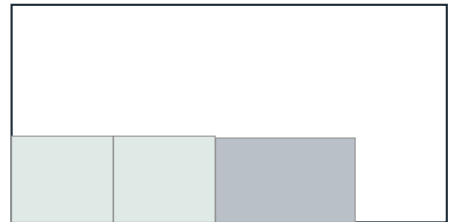
Sheet 10 — 1/4" birch ply 81% used · 7 parts



Sheet 11 — 1/4" birch ply 89% used · 8 parts



Sheet 12 — 1/4" birch ply 31% used · 3 parts



Sheet 13 — 1-9/16" maple slab 75% used · 1 parts

